

Hospital Emergency Queue - Bubble Sort Using Arrays

Real-Time Scenario

CityCare Hospital receives emergency patients. Each patient is assigned an emergency level from 1 (Critical) to 5 (Low).

Input

Patient ID	Name	Level
P101	Rahul	3
P102	Priya	1
P103	Amit	5
P104	Sneha	2
P105	Arjun	4

Diagram

Current Queue

Rahul(3) -> Priya(1) -> Amit(5) -> Sneha(2) -> Arjun(4)

Bubble Sort

Priya(1) -> Sneha(2) -> Rahul(3) -> Arjun(4) -> Amit(5)

Doctors always treat from LEFT to RIGHT.

Pseudo Algorithm

```
START
Read PatientID[]
Read PatientName[]
Read EmergencyLevel[]

FOR each pass
    Compare adjacent levels
    IF left > right
        Swap EmergencyLevel
        Swap PatientID
        Swap PatientName
    END IF
END FOR

Display Queue
STOP
```

Dry Run

```
Initial : 3 1 5 2 4
Pass 1 : 1 3 2 4 5
Pass 2 : 1 2 3 4 5
Sorted Queue Ready
```

Complete Java Solution

```
public class HospitalEmergencyQueue {

    public static void main(String[] args) {

        String[] patientId = {"P101", "P102", "P103", "P104", "P105"};
```

```

String[] patientName = {"Rahul","Priya","Amit","Sneha","Arjun"};
int[] level = {3,1,5,2,4};

for(int i=0;i<level.length-1;i++){
    for(int j=0;j<level.length-i-1;j++){
        if(level[j] > level[j+1]){
            int tempLevel = level[j];
            level[j] = level[j+1];
            level[j+1] = tempLevel;

            String tempId = patientId[j];
            patientId[j] = patientId[j+1];
            patientId[j+1] = tempId;

            String tempName = patientName[j];
            patientName[j] = patientName[j+1];
            patientName[j+1] = tempName;
        }
    }
}

System.out.println("Treatment Queue");

for(int i=0;i<level.length;i++){
    System.out.println(patientId[i]+" "
        +patientName[i]+" Level-"+level[i]);
}
}

```

Expected Output

```

Treatment Queue

P102 Priya Level-1
P104 Sneha Level-2
P101 Rahul Level-3
P105 Arjun Level-4
P103 Amit Level-5

```

Time Complexity: $O(n^2)$

Space Complexity: $O(1)$